

duty, Level-4 ready trucks from Isuzu. Gautam Narang, co-founder and CEO of Gatik, said in a recent interview that they expect to have hundreds of trucks deployed in commercial operations by the end of 2023.

Udelv is another California based company in this space. Its Transporter is a cableless autonomous delivery vehicle with a modular cargo pod that can carry a total of 900kg of goods. The company targets middle- and last-mile delivery. Travelling at speeds up to 112kmph, Udelv's vehicle can cover between 250 and 500km depending on the battery pack, and make up to 80 stops per cycle. Udelv hopes to have 50,000 vehicles operating on public roads by 2028.

In December 2022, Carrefour Drive and Goggo Network collaboratively trialled mobile delivery using an autonomous delivery vehicle operating on public roads, both in heavy traffic and in normal conditions, covering 15km at up to 70kmph. Online orders are packed and loaded on the shuttle, which makes its way to customers who can open their locker using a password shared through SMS to retrieve the goods. The shuttle is capable of making multiple stops in one run. Carrefour has obtained authorisation to run its vehicle on a circuit of 15 kilometres in France.

Since they are heavy and can prove hazardous in case of a malfunction, the self-driving trucks—on highways as well as public roads—are still highly regulated and are permitted only in a few regions across the world. Forbes contributor and trend watcher, Richard Bishop, writes, “2024 appears to be the launch window for commercial driverless on highways, if statements by Aurora and Kodiak play out according to plan. Therefore, 2023 will see a substantial portion of driverless operations in conducting final on-road validation testing. OEMs will keep their launch dates close to the chest, but I'd be very surprised if they made a move sooner. Waymo marches to their own drummer, and if I had to place a bet, I'd



Ottobot 2.0 delivering orders to a traveller at an airport (Courtesy: Ottobot.io)

put my money on them to be the first mover here, maybe even during the coming year.”

Ground delivery bots

This is where ground delivery bots and small vehicles have an advantage. Regulatory bodies that are wary of large trucks are more open to the idea of small autonomous delivery vehicles and bots. These are four- or six-wheeled vehicles or robots that travel on sidewalks or streets to deliver food and parcels to end consumers. Sensors enable them to assess obstacles and safely navigate around people, animals, and objects. They are small, move slowly, and carry small payloads.

Restaurants, grocery stores, and other establishments partner with service providers to use their app based robots for deliveries within a small radius. When a customer places an order, a robot from the network is assigned to pick up their order from their vendor and deliver it to a pinned location. In most cases, the delivery bot has a locker box that contains the items to be delivered. The customer receives a password to open the locker box. Contactless authentication options may also be used.

Autonomous last-mile delivery robots vary in size and function. Some, like Nuro, are as big as a small car, while others are small and cute—like a locker box with wheels. Some move on pavements, and others on roads. Some can do only a single delivery, while

others have modular containers and can manage multiple stops.

Amazon was one of the earliest to enter this field. They trialled a six-wheeled delivery device called Scout in California for almost three years, but they scaled back the project last year.

FedEx had also developed an interesting

same-day delivery robot, Roxo, in collaboration with famous inventor Dean Kamen. A handsome robot, with flashy lights and all the works, it could even climb curbs and steps to deliver packages directly to a customer's door, making it more accessible to customers with limited mobility. However, they shut down the project last year to focus on more near-term opportunities.

Nevertheless, these early movers served as inspiration for a dozen or more startups to make this a very competitive space today!

Nuro was the first company to receive an autonomous exemption from the National Highway Traffic Safety Administration, USA. Its four-wheeled machine weighs nearly 700kg and has a top speed of around 70kmph. It can carry payloads of 225kg, with heating and cooling capabilities and customisable compartments. Nuro has piloted solutions in Texas, Arizona, and California for partners like Dominos, Kroger, Walmart, FedEx, and 7-Eleven. They recently signed a 10-year deal with Uber Eats for robot food delivery.

Not all robots are this big. Those from Starship, Kiwibot, Cyan Robotics, and Ottonomy.IO are small and cute. Kiwibot has deployed over 50 robots in more than 25 universities, and has made more than 200,000 deliveries in the cities of San Jose, Santa Monica, Denver, and Dallas in the US, and in parts of Taiwan and Columbia.